

Manolis PERROT

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Date of birth: April 25, 1997
Nationality: French

Expertise

Physics: Ocean and Atmosphere Turbulence, Subgrid Parameterizations for Climate models, Convection, Energy conservation, Physics-Dynamics coupling.
Maths: Geometric Fluid Dynamics, Deterministic and Stochastic Modeling, Uncertainty Quantification.

Academic experiences

2021 – 2025	PhD research supervised by Eric Blayo , Florian Lemarié at Laboratoire Jean-Kuntzmann , Grenoble: <i>Modeling Oceanic and Atmospheric Convection: Energy, Uncertainties and Rotation</i> .
2020 – 2021 (10 months)	ENS pre-doctoral internship supervised by Eric Blayo , Florian Lemarié and Etienne Mémin at Laboratoire Jean-Kuntzmann , Grenoble: <i>Stochastic modelling of air-sea turbulent fluxes</i> .
2020 (5 months)	M2 Master's thesis (remotely) supervised by Klas Modin (Göteborg Univ., Sweden): <i>Some geometric aspects of ideal hydrodynamics</i> .
2018 (5 months)	Internship supervised by Antoine Venaille and Pierre Delplace at Physics Laboratory of ENS de Lyon: <i>Topological transition in stratified fluids</i> about topological transition (in the sense of condensed matter physics) in stratified and compressible fluids, and origin of Lamb waves in the atmosphere.
2017 (3 months)	Numerical PDE project supervised by Emmanuel Dormy : <i>A model of water waves amplification due to wind stress</i> .
2017 (4 months)	Bachelor's thesis supervised by Laurent Charles : <i>Egorov's theorem and mathematical quantization</i> .
2016 (6 months)	TiPE experiments and thesis for the ENS exam entrance: <i>Electrical resistivity tomography methods applied to the detection of mountain permafrost</i> .

Publications and preprints

Perrot M., Lemarié F. Energetically consistent Eddy-Diffusivity Mass-Flux schemes. Part II: implementation and evaluation in an oceanic context., *under rev.*, JAMES, 2024 [HAL preprint] .
Perrot M., Lemarié F., [Energetically consistent Eddy-Diffusivity Mass-Flux convective schemes. Part I: Theory and Models.](#), *Journal of Advances in Modeling Earth System*, 2025.
Perrot M., Modin K., [Eulerian and Lagrangian stability in Zeitlin's model of hydrodynamics](#), *Communications in Mathematical Physics*, 2024 [arXiv] .
Perrot M., Delplace P., Venaille A. [Topological transition in stratified fluids](#), *Nature Physics* (2019) [arXiv].

Education

2021 – 2025	ECOLE DOCTORALE MSTII, UNIVERSITÉ GRENOBLE ALPES, Grenoble, France. PhD student in applied mathematics: <i>Modeling Oceanic and Atmospheric Convection: Energy, Uncertainties and Rotation</i> .
2016 – 2020	SORBONNE UNIVERSITÉ & ECOLE NORMALE SUPÉRIEURE (ENS) ¹ , Paris, France. Bachelor and Master of pure Mathematics , additional courses in physics and climate sciences.
2014 – 2016	CLASSES PRÉPARATOIRES STANISLAS (PCSI – PC*), Paris, France. Successful scientific preparation for the national entrance exams to the ENS, with major in mathematics, physics and chemistry.

IT skills

TeX, Python, Bash, Fortran, Git, Unix.

Community service

2024 – 2026	Elected member of the Research Commission of Université Grenoble-Alpes.
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Teaching & outreach

¹Selective french school preparing students to research or teaching career. Students are paid as civil servants during their 4-year tuition

2021 – 2022	Teaching assistant in linear algebra for civil engineering students (60h).
2017	Organization of a workshop during a meeting for highschool girls in mathematics (Rendez-vous des Jeunes Mathématiciennes).
2016 – 2018 (2 years)	Oral examiner (<i>colleur</i>) in <i>Classe préparatoire</i> in Physics.

Languages

French:	Mother tongue.
English:	Fluent.
Spanish:	Fluent.
Portuguese:	Good working knowledge.

Events and Talks

October 2024	Speaker, Fine Scales in Numerical Ocean Models workshop, Brest, France: <i>Eddy-Diffusivity Mass-Flux convective parameterization: energy budgets, rotation effects and uncertainty quantification</i> .
May 2024	GDR DEPHY - <i>Development and Evaluation of Physical Parameterizations</i> workshop, Fréjus, France .
April 2024	Speaker, EGU Vienna, Austria: <i>Convective Mixing: an Energetically Consistent Non-Local Parameterization</i> .
March 2024	Invited speaker, PC6-7 TRACCS kickoff meeting Toulouse, France: <i>EDMF convective schemes: closed energy budgets and transport-map Bayesian estimation</i> .
January 2024	Speaker, Ateliers de Modélisation de l'Atmosphère Toulouse, France: <i>Energetically consistent coupling of EDMF and TKE schemes for Convection</i> .
December 2023	Invited speaker, MathsInFluid seminar ENS de Lyon, France: <i>Au-delà de la diffusion turbulente: la convection et ses paramétrisations</i> .
September 2023	Fluid Dynamics of Sustainable Environment summer school , Cambridge University, United-Kingdom.
July 2023	<i>Derivation and calibration of consistent Eddy-Diffusivity Mass-Flux parameterizations for ocean models</i> , talk at the PDEs on the Sphere conference , Grenoble, France .
June 2023	Attendee of Journée annuelle du GDR Théorie et Climat , Paris, France .
May 2023	GDR DEPHY - <i>Development of Physical Parameterizations</i> workshop, Fréjus, France .
April 2022	Short talk at the research school <i>Mathematical Advances in Geophysical Flows</i> , CIRM, Marseille, France : Multi-fluid modelling for Ocean Deep Convection Parameterizations.
November 2021	Attendee of the winter school <i>Numerical modelling of Ocean and Atmosphere</i> , Grenoble, France.
August 2021	Attendee of the <i>Workshop on Ocean-Atmosphere interaction</i> , Ecole de Physique des Houches, France.
June 2018	Attendee of the <i>Mathematical Physics Days : Quantum Chaos</i> , Lyon, France.